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Course code: Name CSAD301 - DATA STRUCTURES

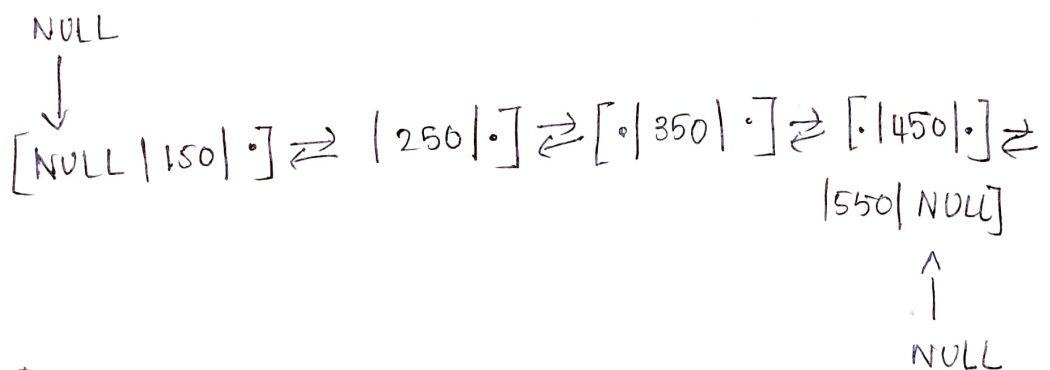
COA - Assessment Tool 1: Analytical Problems

- 1) Insert the following elements 150, 250, 350, 450, 550 into the double linked list and delete the elements 150 and 250 from the double linked list. Show the linked structure using a node diagram before and after delete operations

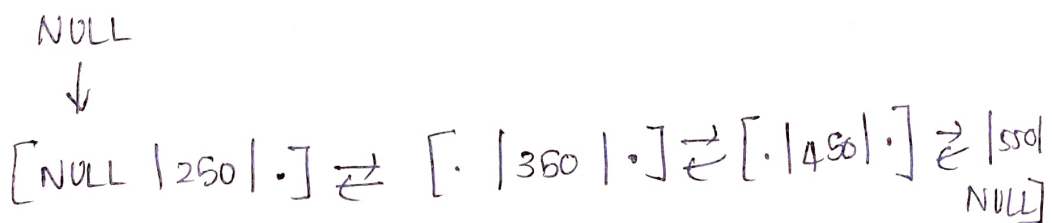
Doubly linked list

Insertion of elements: 150, 250, 350, 450, 550

Node diagram before deletion:

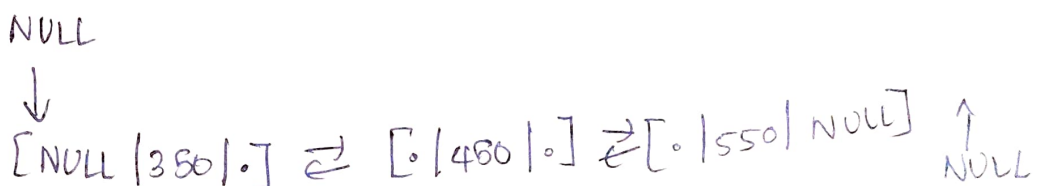


Delete 150



Delete 250

Final linked structure



Final elements: $350 \Rightarrow 450 \Rightarrow 550$

- 2) A circular queue has a size of 6. Execute the following operations: Enqueue(10), Enqueue(20), Enqueue(30), Dequeue(), Enqueue(40), Enqueue(50), Enqueue(60), Enqueue(70). Draw the queue after each operation & identify the values of the front and rear pointers.

Circular Queue of size 6

Assume array positions are indexed from 0 to 5

Initially: Front = -1, Rear = -1

Enqueue: $\text{Rear} = (\text{Rear} + 1) \% 6$

Dequeue: $\text{Front} = (\text{Front} + 1) \% 6$

Operation 1: Enqueue(10)

[10] [] [] [] [] []

↑

Front = 0

↑

Rear = 0

Operation 2: Enqueue (20)

[10] [20] [] [] []
↑ ↑
Front Rear

Front = 0

Rear = 1

Operation 3: Enqueue (30)

[10] [20] [30] [] [] []
↑ ↑
Front Rear

Front = 0

Rear = 2

Operation 4: Dequeue()

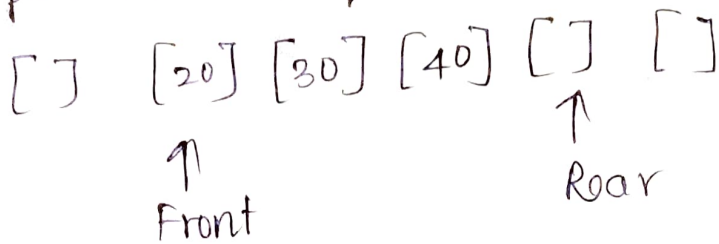
10 is removed

[] [20] [30] [] [] []
 ↑ ↑
 Front Rear

Front = 1

Rear = 2

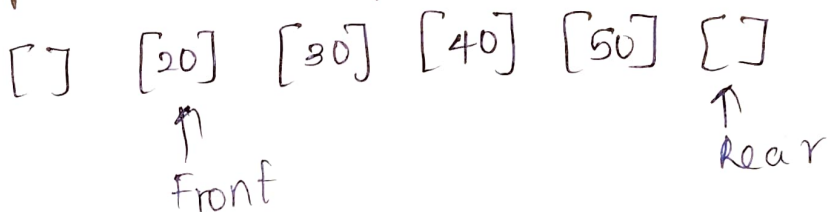
Operation 5: Enqueue (40)



Front = 1

Rear = 3

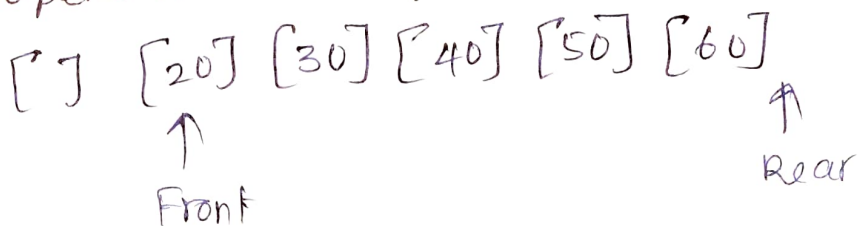
Operation 6: Enqueue (50)



Front = 1

Rear = 4

Operation 7: Enqueue (60)



Front = 1

Rear = 5

Operation 8: Enqueue (70)

Since the queue is circular, the rear wraps around:

wraps around

[70] [20] [30] [40] [50] [60]

↑ Rear ↑ Front

Front = 1

Rear = 0

Final Circular Queue:

Front $\rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow 50 \rightarrow 60 \rightarrow 70 \rightarrow$ (back to Front)
 $\uparrow$ $\uparrow$
 position 1 position 0

Final values :

Front = 1 $\rightarrow$ 20

Rear = 0 $\rightarrow$ 70

Queue is FULL because all 6 positions are occupied.